

Product Manual

Smart Eddy Cirqus Voltaire

P/N TPP-1040-CV

Document Revision – 1.0



The 'Smart Eddy – Cirqus Voltaire' (Part Number TPP-1040-CV) board is a variant of the popular Tangles Smart Eddy designed for use in the game Cirqus Voltaire. It is a modern re-design of the original part number A-22149-1.

Replace your missing, broken or unreliable A-22149-1 boards with TPP-1040 to revive the phantom flip feature. The TPP-1040 will give a reliable long service to your MB machine.

The TPP-1040 is drop-in compatible with the original part, using the same mounting holes. The board is supplied with 4 new nylon mounting posts.

Installation is a simple matter of installing the board with the 4 original screws and connecting the 2-wire sensor cable to J1 and the 4-wire power/signal cable to J2.

The Smart Eddy boards utilise an auto-calibration feature. This feature is activated each time the machine is turned on.

LED SIGNALS

When the board is powered on it should display a quick flash of the LED. The LED should come on solid for 2 to 4 seconds (calibration phase). Finally the board will give a quick blink of the LED and then go OFF.

1. Quick Flash – Sensor coil detected
2. 1 to 4 seconds on – Calibrating
3. Blink 3 times – Calibration successful
4. LED Off – Ready for gameplay.

TROUBLESHOOTING

LED Does not come on

Check J2. Check fuse for unregulated 12v supply.

LED Comes on Solid (Never blinks)

This indicated the sensor has not been detected. Check connector J1. Check sensor 2 pin plug (can lose their tension and not make good contact. Check sensor inductor in the playfield routed slot.

Poor quality or worn-out 2-pin Molex connections can make the operation of the eddy board unreliable. It is recommend you check the 2-pin Molex has a good “grip” on the pins of the Smart Eddy PCB. Crimping a new 2-Pin molex onto the sensor is not a difficult task and can solve many problems.

See Molex Parts:

22-01-3027 (KK 254 connector crimp housing)

08-50-0032 (KK 254 cat ear crimp terminal)

Powerup LED

When the Smart Eddy board has power applied, it will perform a series of LED blinks. The boot-up sequence is:

1. Quick Flash – Sensor coil detected.
2. 1 to 4 seconds on – Calibrating.
3. Blink 3 times – Calibration successful.
4. LED Off – Ready for gameplay

Detect LED

With the game powered up, you can pass a pinball over the detection zone on the playfield and look to see if the Smart-Eddy board LED comes on when the ball is near the sensor.

If the light fails to illuminate at all, it could be a sign there is no 12v power getting to the Smart Eddy. Check Power driver board F101.

Switch Test

By powering the game up and entering into Switch-Test mode, you can verify the switch matrix connections to the Smart Eddy are working.

Smart Eddy Ground Connection

The CV game has control over the power to the Eddy board through solenoid #40 (J110 pin 5).

The ground passed to the smart eddy is not a true ground, but under the control of the CPU. The CPU can remove the ground from the eddy board forcing it to reset. If this action is occurring too often I recommend one of 2 actions:

- Remove the BLUE-WHITE wire from J110-5 and connect the wire to the game ground.
- Install ROM Revision 2.0h

The Smart-Eddy has an automatic “lockup” detection built in and can re-calibrate mid game if a detection lockup occurs. CPU initiated resets of the Smart-Eddy board are not necessary.

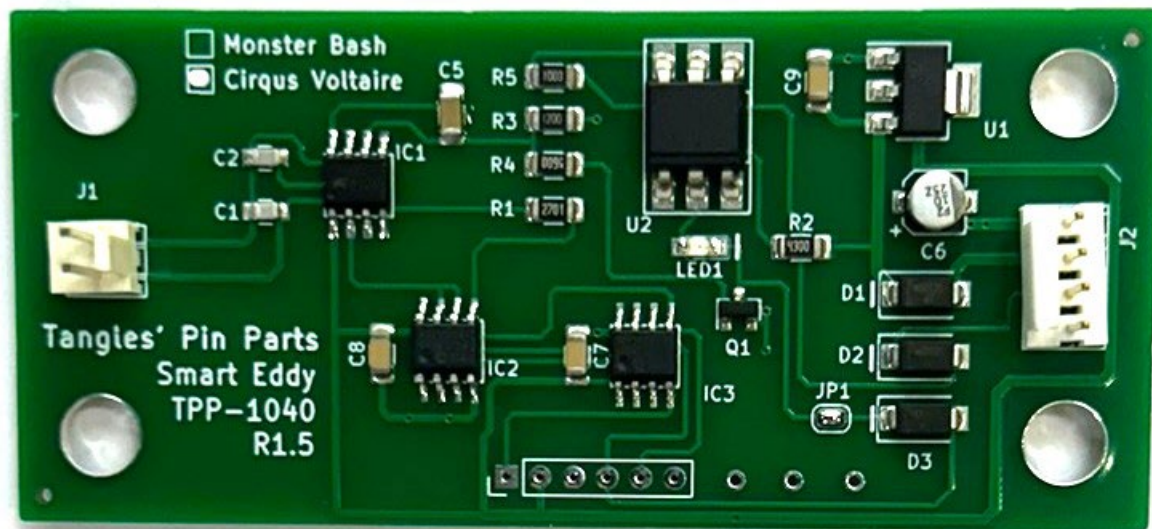
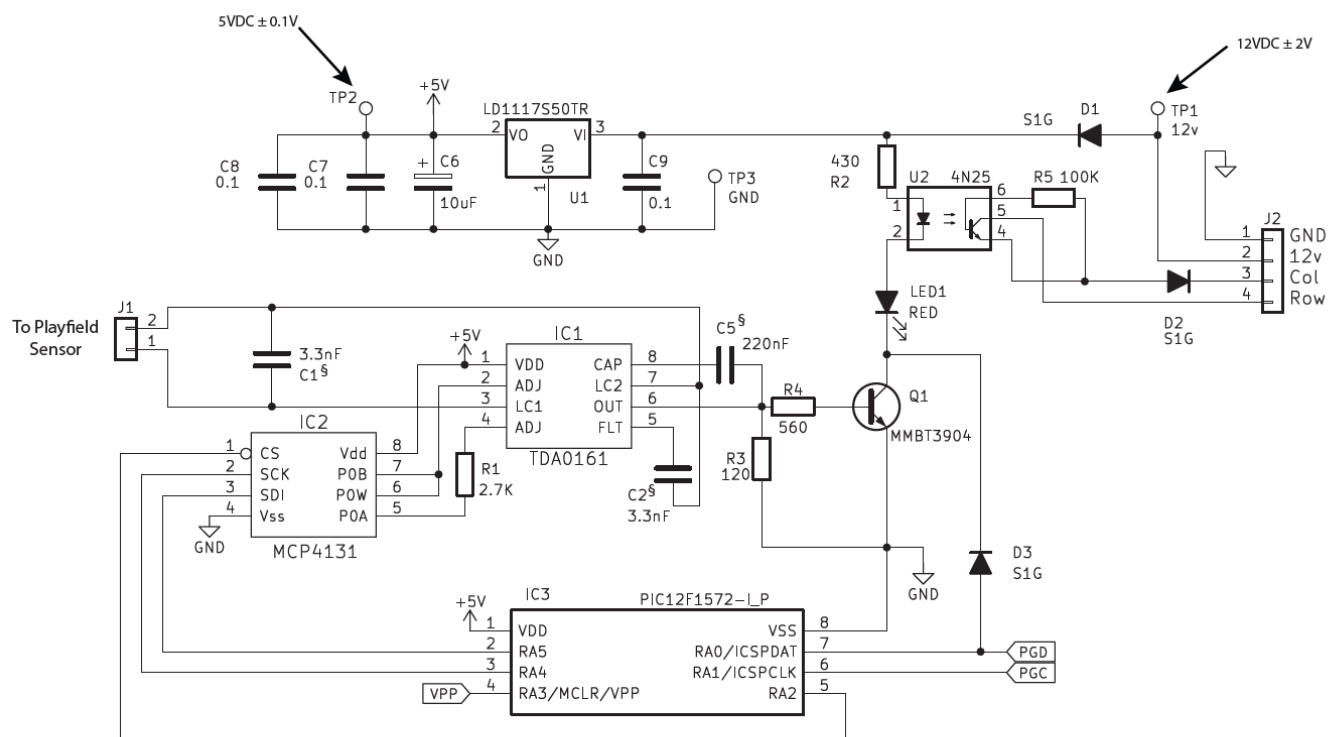


Photo 1



§ Circus Voltaire

C1 = 680pF
C2 = 820pF
C5 = 10nF

Parts List

Designator	Description
U1	LD1117S50TR
U2	4N25
IC1	TDA0161
IC2	MCP4131-103
IC3	PIC12F1572-LP
Q1	BC817
C1	680pF
C2	820pF
C5	10nF
C6	10uF 25v
C7,C8,C9	100nF
D1,D2,D3	S1G
R1	4.7KΩ
R2,R4	560Ω
R3	120Ω
R5	100KΩ

